AS3504 Exam #1 Study Guide

15 True/False questions (1 point each) + 5 Numerical Free Response questions (5 points each)

75 Minutes, 40 Points

Key: Understand terminology & notation so you can excel at word problems.
ie. There are many ways to say predictors, MSE, error sum of squares, etc.

The listed topics are for numerical Qs. Conceptual Qs will show up from any section.

Statistical Learning (1 Q):
- Bayes Classification Error
- K-Nearest Neighbors

Simple Linear Regression (2 Qs):
- Least squares coefficient estimation and R^2
- Model predictions, calculator usage
- Residuals, MSE, SST, SSE, SSReg, R^2 calculations

Simple Linear Regression Inference (2 Qs):
- t-statistic, hypothesis tests, p-values
- Variance/standard error of coefficient estimates
- Confidence/Prediction Intervals

Vocabulary to Know:
- Response vs. Explanatory variable
- Regression vs. Classification
- Prediction vs. Inference
- Training vs. Test data
- Total Sum of Squares vs. Error Sum of Squares (aka. Residual Sum of Squares) vs. Regression Sum of Squares and relationship with Residual, Mean Squared Error, Residual Standard Error, R^2
- Mean Squared Error vs. Residual Standard Error
- Stochastic vs. Non-stochastic variable
- Heteroscedastic vs. Homoscedastic
- Null vs. Alternate Hypothesis
- Variance vs. Standard Error
- p-value vs. critical α vs. t-statistic and relationship w/hypothesis testing and conclusion
- Confidence vs. Prediction interval

Formulas to Remember:

- Bayes Error Rate
- Distance Formula for KNN
- $b_0 = \bar{y} - b_1\bar{x}$
- $b_1 = r\frac{s_y}{s_x} = \frac{Cov(X,Y)}{s_x^2} = \frac{\sum_{i=1}^n (x_i-\bar{x})(y_i-\bar{y})}{\sum_{i=1}^n (x_i-\bar{x})^2} = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2}$ (will be provided if needed)
- Residual: $e_i = y_i - \hat{y}_i$
- $SST = SSE + SSReg$
- $SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$
- $SST = \sum_{i=1}^n (y_i - \bar{y})^2$
- $SSReg = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = b_1^2 \sum_{i=1}^n (x_i - \bar{x})^2 = b_1 \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$
- $MSE = s^2 = \frac{SSE}{n-2}$
- $s_y^2 = \frac{SST}{n-1}$
- $r = \frac{Cov(X,Y)}{s_x s_y}$ and using your calculator
- $R^2 = \frac{SSReg}{SST}$
- $se(b_0) = s\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i-\bar{x})^2}}$
- $se(b_1) = \frac{s}{\sqrt{\sum_{i=1}^n (x_i-\bar{x})^2}}$
- $t = \frac{Estimator - H_0\ value}{se(Estimator)}$
- $F = \frac{MSReg}{MSE} = t^2$ and for SLR, $F = \frac{SSReg}{MSE}$
- Confidence Interval: $b_1 \pm t_{n-2,\frac{\alpha}{2}} * se(b_1)$
- $se(\hat{y}_{pred}) = s\sqrt{1 + \frac{1}{n} + \frac{(x_{pred}-\bar{x})^2}{\sum_{i=1}^n (x_i-\bar{x})^2}}$
- Prediction Interval: $\hat{y}_{pred} \pm t_{n-2,\frac{\alpha}{2}} * se(\hat{y}_{pred})$